Module 11 : Block randomized experiments

Marie-Abèle Bind

Design of Experiments - Stat 140

Spring 2026

Randomized block experiments

- Why block (also called stratify) ?
- E.g., demographic characteristics [sex differences]
- Fisher's rule : "Block what you can, randomize what you cannot".
- Sometimes difficult, why ?
- Newer method that requires computational capabilities : rerandomization.

Taxonomy of randomized block experiments

- In order to ensure that each treatment is assigned to at least one experimental unit within each block, the size of each block needs to be at least the number of treatment levels.
- If this condition is satisfied, the design is called "complete block design"; otherwise, "incomplete block design".
- A design is said to be balanced (in size) if the number of experimental units assigned to each treatment within each block is the same.
- Balanced complete block designs have some attractive properties (e.g., orthogonal decomposition of the total sum of squares into different components).
- Often possible in engineering and industrial settings, less so in social and biomedical experiments.

Our setting (Chapter 9, Section 9.3.1.)

- A single factor with two levels is applied.
- One blocking variable with two levels.
  - f : female and m : male
- We consider a population of N experimental units.
  - N=N(f)+N(m)

Assignment mechanism

- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y(0)},\mathbf{Y(1)}) = \begin{cases} \binom{N(f)}{N_t(f)}^{-1}\binom{N(m)}{N_t(m)}^{-1} & \text{for } \mathbf{W} \in \mathbb{W}^+ \\ 0 & \text{otherwise} \end{cases}$$

where

$$\mathbb{W}^+ = \left\{ \mathbf{W} \text{ such that } \sum_{i \in \text{Block } f} W_i = N_t(f) \text{ and } \sum_{i \in \text{Block } m} W_i = N_t(m) \right\}$$

Science Table

- Let $Y_i(W_i = 0)$ and $Y_i(W_i = 1)$ denote the ith unit's potential outcomes under control and treatment, respectively.

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Causal estimands

Science Table

- Let $Y_i(W_i = 0)$ and $Y_i(W_i = 1)$ denote the ith unit's potential outcomes under control and treatment, respectively.

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Causal estimands
  - Unit-level causal effects of treatment versus control within block k :

$$\tau_i(k) = Y_i(W_i = 1) - Y_i(W_i = 0) \quad \text{for } i \in \text{Block } k$$

Science Table

- Let $Y_i(W_i = 0)$ and $Y_i(W_i = 1)$ denote the ith unit's potential outcomes under control and treatment, respectively.

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Causal estimands
  - Average causal effect of treatment versus control within block k :

$$\tau(k) = \frac{1}{N(k)} \sum_{i \in \text{Block } k} \tau_i(k)$$

Science Table

- Let $Y_i(W_i = 0)$ and $Y_i(W_i = 1)$ denote the ith unit's potential outcomes under control and treatment, respectively.

Unit i	Block k	$\mathrm{Y}_i(W_i = 0)$	$\mathrm{Y}_i(W_i = 1)$
1	f	$\mathrm{Y}_1(W_i = 0)$	$\mathrm{Y}_1(W_i = 1)$
2	f	$\mathrm{Y}_2(W_i = 0)$	$\mathrm{Y}_2(W_i = 1)$
3	m	$\mathrm{Y}_3(W_i = 0)$	$\mathrm{Y}_3(W_i = 1)$
...	...	...	...
N	m	$\mathrm{Y}_N(W_i = 0)$	$\mathrm{Y}_N(W_i = 1)$

- Causal estimands
  - Average causal effect of treatment versus control :

$$\tau = \frac{N(f)}{N(f) + N(m)}\,\tau(f) + \frac{N(m)}{N(f) + N(m)}\,\tau(m)$$

Science Table

- Let $Y_i(W_i = 0)$ and $Y_i(W_i = 1)$ denote the ith unit's potential outcomes under control and treatment, respectively.

Unit i	Block k	$\mathrm{Y}_i(W_i = 0)$	$\mathrm{Y}_i(W_i = 1)$
1	f	$\mathrm{Y}_1(W_i = 0)$	$\mathrm{Y}_1(W_i = 1)$
2	f	$\mathrm{Y}_2(W_i = 0)$	$\mathrm{Y}_2(W_i = 1)$
3	m	$\mathrm{Y}_3(W_i = 0)$	$\mathrm{Y}_3(W_i = 1)$
...	...	...	...
N	m	$\mathrm{Y}_N(W_i = 0)$	$\mathrm{Y}_N(W_i = 1)$

- Causal estimands

- Note that, for each unit i, only one of the two potential outcomes can be observed.

Observed Table

Unit i	Block k	W_i	$\mathrm{Y}_i(W_i = 0)$	$\mathrm{Y}_i(W_i = 1)$
1	f	0	$\mathrm{Y}_1(W_i = 0)$	?
2	f	1	?	$\mathrm{Y}_2(W_i = 1)$
3	m	1	?	$\mathrm{Y}_3(W_i = 1)$
...	...	...	...	...
...	...	...	...	...
N	m	0	$\mathrm{Y}_N(W_i = 0)$	?

- How to write Y_i^{obs} ?

Observed Table

Unit i	Block k	W_i	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	0	$Y_1(W_i = 0)$	?
2	f	1	?	$Y_2(W_i = 1)$
3	m	1	?	$Y_3(W_i = 1)$
...	...	...	...	...
...	...	...	...	...
N	m	0	$Y_N(W_i = 0)$	?

- How to write Y_i^{obs} ?

$$Y_i^{obs} = W_i Y_i(W_i = 1) + (1 - W_i) Y_i(W_i = 0)$$

Fisherian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Sharp null hypothesis

Fisherian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Test statistic

Fisherian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Test statistic
  - $\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)$
  - $\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)$

Fisherian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Test statistic
  - $\lambda\ [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] + (1-\lambda)\ [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$

    for some $\lambda \in [0, 1]$.

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

Neymanian inference

Unit i	Block k	$\mathrm{Y}_i(W_i = 0)$	$\mathrm{Y}_i(W_i = 1)$
1	f	$\mathrm{Y}_1(W_i = 0)$	$\mathrm{Y}_1(W_i = 1)$
2	f	$\mathrm{Y}_2(W_i = 0)$	$\mathrm{Y}_2(W_i = 1)$
3	m	$\mathrm{Y}_3(W_i = 0)$	$\mathrm{Y}_3(W_i = 1)$
...	...	...	...
N	m	$\mathrm{Y}_N(W_i = 0)$	$\mathrm{Y}_N(W_i = 1)$

- Estimator for $\tau(k)$?

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Estimator for $\tau(k)$?
  - $\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k)$

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Estimator for $\tau(k)$?
  - $\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k)$
- Sampling variance of $\hat{\tau}(k)$:

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Estimator for $\tau(k)$?
  - $\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k)$
- Sampling variance of $\hat{\tau}(k)$:
  - $Var(\hat{\tau}(k)) = \frac{S_T^2(k)}{N_T(k)} + \frac{S_C^2(k)}{N_C(k)} - \frac{S_{TC}^2(k)}{N(k)}$

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Estimator for $\tau(k)$?
  - $\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k)$
- Sampling variance of $\hat{\tau}(k)$:
  - $Var(\hat{\tau}(k)) = \frac{S_T^2(k)}{N_T(k)} + \frac{S_C^2(k)}{N_C(k)} - \frac{S_{TC}^2(k)}{N(k)}$
- Estimator for the sampling variance of $\hat{\tau}(k)$:

Neymanian inference

Unit i	Block k	$Y_i(W_i = 0)$	$Y_i(W_i = 1)$
1	f	$Y_1(W_i = 0)$	$Y_1(W_i = 1)$
2	f	$Y_2(W_i = 0)$	$Y_2(W_i = 1)$
3	m	$Y_3(W_i = 0)$	$Y_3(W_i = 1)$
...	...	...	...
N	m	$Y_N(W_i = 0)$	$Y_N(W_i = 1)$

- Estimator for $\tau(k)$?
  - $\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k)$
- Sampling variance of $\hat{\tau}(k)$:
  - $Var(\hat{\tau}(k)) = \frac{S_T^2(k)}{N_T(k)} + \frac{S_C^2(k)}{N_C(k)} - \frac{S_{TC}^2(k)}{N(k)}$
- Estimator for the sampling variance of $\hat{\tau}(k)$:
  - $\hat{Var}(\hat{\tau}(k)) = \frac{s_T^2(k)}{N_T(k)} + \frac{s_C^2(k)}{N_C(k)}$

    [Assuming constant and additive $\tau_i(k)$]

Neymanian inference

- Estimator for τ ?

Neymanian inference

- Estimator for τ ?
  - $\hat{\tau} = \frac{N(f)}{N(f)+N(m)} \, [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] + \frac{N(m)}{N(f)+N(m)} \, [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$

Neymanian inference

- Estimator for τ ?
  - $\hat{\tau} = \frac{N(f)}{N(f)+N(m)} \left[\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)\right] + \frac{N(m)}{N(f)+N(m)} \left[\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)\right]$
- Sampling variance of $\hat{\tau}$:

Neymanian inference

- Estimator for τ ?
  - $\hat{\tau} = \frac{N(f)}{N(f)+N(m)} \, [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] + \frac{N(m)}{N(f)+N(m)} \, [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$
- Sampling variance of $\hat{\tau}$:
  - $$Var(\hat{\tau}) = (\frac{N(f)}{N(f)+N(m)})^2 \, [\frac{S_T^2(f)}{N_T(f)} + \frac{S_C^2(f)}{N_C(f)} - \frac{S_{TC}^2(f)}{N(f)}] + (\frac{N(m)}{N(f)+N(m)})^2 \, [\frac{S_T^2(m)}{N_T(m)} + \frac{S_C^2(m)}{N_C(m)} - \frac{S_{TC}^2(m)}{N(m)}]$$

Neymanian inference

- Estimator for τ ?
  - $\hat{\tau} = \frac{N(f)}{N(f)+N(m)} \; [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] + \frac{N(m)}{N(f)+N(m)} \; [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$
- Sampling variance of $\hat{\tau}$:
  - $$\begin{aligned} Var(\hat{\tau}) = & \left(\frac{N(f)}{N(f)+N(m)}\right)^2 \left[\frac{S_T^2(f)}{N_T(f)} + \frac{S_C^2(f)}{N_C(f)} - \frac{S_{TC}^2(f)}{N(f)}\right] \\ & + \left(\frac{N(m)}{N(f)+N(m)}\right)^2 \left[\frac{S_T^2(m)}{N_T(m)} + \frac{S_C^2(m)}{N_C(m)} - \frac{S_{TC}^2(m)}{N(m)}\right] \end{aligned}$$
- Estimator for the sampling variance of $\hat{\tau}$:

Neymanian inference

- Estimator for τ ?
  - $\hat{\tau} = \frac{N(f)}{N(f)+N(m)} \; [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] + \frac{N(m)}{N(f)+N(m)} \; [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$
- Sampling variance of $\hat{\tau}$:
  - $$\begin{aligned} Var(\hat{\tau}) = & \left(\frac{N(f)}{N(f)+N(m)}\right)^2 \left[\frac{S_T^2(f)}{N_T(f)} + \frac{S_C^2(f)}{N_C(f)} - \frac{S_{TC}^2(f)}{N(f)}\right] \\ & + \left(\frac{N(m)}{N(f)+N(m)}\right)^2 \left[\frac{S_T^2(m)}{N_T(m)} + \frac{S_C^2(m)}{N_C(m)} - \frac{S_{TC}^2(m)}{N(m)}\right] \end{aligned}$$
- Estimator for the sampling variance of $\hat{\tau}$:
  - $\hat{Var}(\hat{\tau}) = \left(\frac{N(f)}{N(f)+N(m)}\right)^2 \left[\frac{s_T^2(f)}{N_T(f)} + \frac{s_C^2(f)}{N_C(f)}\right] + \left(\frac{N(m)}{N(f)+N(m)}\right)^2 \left[\frac{s_T^2(m)}{N_T(m)} + \frac{s_C^2(m)}{N_C(m)}\right]$

  [Assuming $S_{TC}^2(k) = 0 \;\; \forall k$]

Other estimand : $\tau(m) - \tau(f)$

- Unbiased estimator for $\tau(m) - \tau(f)$:
  - $\hat{\tau}(m) - \hat{\tau}(f) = [\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)] - [\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)]$
- Sampling variance of $\hat{\tau}(m) - \hat{\tau}(f)$:
  - $\text{Var}(\hat{\tau}(m) - \hat{\tau}(f)) = \frac{S_T^2(f)}{N_T(f)} + \frac{S_C^2(f)}{N_C(f)} - \frac{S_{TC}^2(f)}{N(f)} + \frac{S_T^2(m)}{N_T(m)} + \frac{S_C^2(m)}{N_C(m)} - \frac{S_{TC}^2(m)}{N(m)}$
- Estimator for the sampling variance of $\hat{\tau}(m) - \hat{\tau}(f)$:
  - $\hat{Var}(\hat{\tau}(m) - \hat{\tau}(f)) = \frac{s_T^2(f)}{N_T(f)} + \frac{s_C^2(f)}{N_C(f)} + \frac{s_T^2(m)}{N_T(m)} + \frac{s_C^2(m)}{N_C(m)}$